Problem F. Equations

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2.5 seconds
Memory limit:	256 megabytes

Let $f(a, b, m)$ be the **least non-negative** solution of the congruence equation:

$$ax \equiv b \pmod{m}$$

If there isn't a solution, then $f(a, b, m) = 0$.

Given n, a, b, calculate $\sum_{i=1}^{n} f(a, b, i) \bmod 998244353$.

Input

The first line contains an integer T ($1 \le T \le 5$) denoting the number of test cases.

Then T lines follow, each containing three positive integers n, a, b ($1 \le n \le 10^{18}$, $1 \le a, b \le 10^6$).

Output

Output T lines. The i-th line contains the answer of the i-th test case.

Example

standard input	standard output
5 5 4 3 5 3 4 10 5 8 10 8 5 100 79 97	2 3 15 10 2519